

STATISTICS FOR DATA ANALYSTS

TUESDAY

Probability & Normal Distribution

Today's Big Questions:

1. How likely is something to happen? (PROBABILITY)
2. What shape does most real-world data follow? (NORMAL DISTRIBUTION)

Why This Matters:

Understanding probability helps you make predictions and assess risk. Understanding normal distribution helps you identify unusual patterns in data and apply statistical tests. These two concepts are the foundation for everything that comes next in statistics.

By the end of today, you will be able to:

- ✓ Calculate the probability of events
- ✓ Understand what the 68-95-99.7 rule means (and use it!)
- ✓ Identify if your data is normally distributed
- ✓ Find outliers using the normal distribution
- ✓ Use z-scores to standardize data

1. PROBABILITY — How Likely Is Something?

Definition in Plain English:

Probability is a number that tells you how likely something is to happen. It's always between 0 (impossible) and 1 (certain to happen). You can also express it as a percentage (0% to 100%).

The Most Basic Example:

Flip a fair coin. Two outcomes are possible: Heads or Tails. Each has an equal chance.

- Probability of Heads = 1 out of 2 = 0.5 = 50%
- Probability of Tails = 1 out of 2 = 0.5 = 50%

Another Simple Example: Rolling a Die

A six-sided die has numbers 1, 2, 3, 4, 5, 6. If you roll it once:

- Probability of getting a 4 = 1 out of 6 = 0.1667 = 16.67%
- Probability of getting an even number (2, 4, or 6) = 3 out of 6 = 0.5 = 50%
- Probability of getting a number greater than 3 = 3 out of 6 = 0.5 = 50%

The Formula:

$$P(\text{Event}) = (\text{Number of favorable outcomes}) / (\text{Total possible outcomes})$$

Real Data Analyst Examples:

Example 1: Customer Churn

Your churn model says: 'This customer has 0.15 probability of leaving.' That means 15% chance they will churn. Or: if you picked 100 customers like this one, roughly 15 would leave.

Example 2: Fraud Detection

Your fraud detector says: 'This transaction has probability 0.92 of being fraudulent.' That's 92% likely to be fraud. Very suspicious!

Example 3: Marketing

You A/B test an email. The treatment group has probability 0.35 of clicking. That means 35% of people who receive this email will click on it.

Python Code Example:

```
import random # Simulate flipping a coin 100 times
coin_flips = [random.choice(['Heads', 'Tails']) for _ in range(100)]
# Count heads
heads_count = coin_flips.count('Heads')
probability_heads = heads_count / 100
print(f"Out of 100 flips, got {heads_count} heads")
print(f"Probability = {probability_heads}")
# Should be close to 0.5!
# Rolling a die 100 times
die_rolls = [random.randint(1, 6) for _ in range(100)]
sixes = die_rolls.count(6)
probability_six = sixes / 100
print(f"Probability of rolling a 6: {probability_six}")
# Should be close to 0.1667 (1/6)
```

Important: The Three Rules of Probability

Rule	Formula	Meaning	Example
P(A AND B)	$P(A) \times P(B)$	Both events happen (if independent)	Coin shows Heads AND die shows 6 = 0.5×0.1667
P(A OR B)	$P(A) + P(B)$	Either event happens (if mutually exclusive)	Die shows 1 OR 2 = $0.1667 + 0.1667$
P(A B)	Conditional Probability	Probability of A GIVEN B happens	Prob of rain TODAY given it rained YESTERDAY

Common Mistakes Students Make:

Mistake 1: 'If a coin landed heads 5 times in a row, tails is more likely next time.'

Reality: Each flip is independent. The probability is still 50% for heads.

Mistake 2: Confusing 'probability' with 'guarantee.'

Reality: 70% probability means it's likely, but not certain. It could still not happen.

Mistake 3: Thinking rare events can't happen.

Reality: Even 1% probability events happen 1 out of 100 times!

WHY Does Probability Matter in Data Analytics?

Every model you build outputs a probability. Every prediction comes with uncertainty. Every experiment has randomness. Probability helps you quantify that uncertainty and make better decisions.

WHERE Do You Use Probability in Your Job?

- **Churn models:** Predict probability customer will leave
- **Fraud detection:** Probability this transaction is fraudulent
- **Recommendation systems:** Probability user will like this product
- **A/B testing:** Probability that version B is better than A
- **Forecasting:** Probability sales will fall in this range

WHAT DOES IT SHOW?

Probability shows uncertainty. It lets you quantify 'how sure' you are about something. Instead of saying 'maybe,' you say '30% likely' or '85% confident.'

⇒ ■ Practice Problems (Probability):

Problem 1: You have a deck of 52 playing cards. What is the probability of drawing an Ace? (There are 4 Aces in a deck)

Problem 2: In your customer database, 15% are repeat buyers. What is the probability that a random customer is a repeat buyer?

Problem 3: You flip a fair coin twice. What is the probability of getting 2 heads in a row? (Hint: Use the AND rule)

2. NORMAL DISTRIBUTION — The Bell Curve

Definition:

The Normal Distribution is a pattern that appears **CONSTANTLY** in nature and real-world data. It's shaped like a bell, which is why it's called the 'bell curve.' Most values cluster around the middle, and fewer values appear at the extremes.

Why You Encounter It Everywhere:

- **Heights of people:** Most people are around 5'6". Very few are 4'8" or 6'4".
- **Test scores:** Most students score around the average. Few get 0% or 100%.
- **Product weights:** A cereal box says '500g.' Most boxes are 498-502g. Rarely 450g or 550g.
- **Customer spending:** Most customers spend near the average. Few spend nothing, few spend 10x the average.

Key Properties of Normal Distribution:

- ✓ **Symmetric:** Looks the same on both sides of the middle
- ✓ **Bell-shaped:** Highest in the middle, slopes down on both sides
- ✓ **Mean = Median = Mode:** The middle point is all three measures at once
- ✓ **Defined by two numbers:** Mean (where it's centered) and Standard Deviation (how spread out it is)

The Most Important Rule: 68-95-99.7

This rule tells you **HOW MUCH** of the data falls at different distances from the mean. This is the **MOST** powerful tool in statistics.

Distance from Mean	Percentage of Data Inside	Interpretation
Within 1 SD	68%	More than 2 out of 3 values
Within 2 SD	95%	Roughly 19 out of 20 values
Within 3 SD	99.7%	Almost everything (only 0.3% outside)

Visual Representation (Text Description):

Imagine a bell curve centered at the middle (mean). At the **PEAK** (center): Most of your data
Draw a line 1 step left and 1 step right from center = 68% of data inside
Draw a line 2 steps left and 2 steps right from center = 95% of data inside
Draw a line 3 steps left and 3 steps right from center = 99.7% of data inside
The far extremes (beyond 3 steps) = only 0.3% of data (very rare!)

Real Example: Daily Sales

Your company's daily sales: Mean = Rs 50,000, SD = Rs 5,000

Using the 68-95-99.7 rule:

- 68% of days: Rs 45,000 to Rs 55,000 (Mean $\pm$ 1 SD)
- 95% of days: Rs 40,000 to Rs 60,000 (Mean $\pm$ 2 SD)
- 99.7% of days: Rs 35,000 to Rs 65,000 (Mean $\pm$ 3 SD)
- If you see a day with Rs 20,000 sales (4 SD below mean): **ALERT! Something is very wrong!**

Z-Score: Standardizing Data

A z-score tells you HOW MANY STANDARD DEVIATIONS away from the mean a value is.

Formula: $z = (\text{value} - \text{mean}) / \text{SD}$

Example: Daily sales mean = Rs 50,000, SD = Rs 5,000. A day with Rs 55,000 sales:

$$z = (55,000 - 50,000) / 5,000 = 5,000 / 5,000 = 1$$

This means Rs 55,000 is exactly 1 SD above the mean ($z = 1$).

Interpreting Z-Scores:

- $z = 0$: Exactly at the mean (average day)
- $z = 1$ or $z = -1$: One SD away (normal variation)
- $z = 2$ or $z = -2$: Two SDs away (uncommon)
- $z = 3$ or $z = -3$: Three SDs away (very rare!)
- $|z| > 3$: Outlier! This value is extremely unusual.

Python Code: Z-Score & Normal Distribution

```
import numpy as np from scipy.stats import norm # Example data data = [45000, 48000, 50000, 52000, 55000, 60000, 35000] mean = np.mean(data) sd = np.std(data) # Calculate z-scores z_scores = [(x - mean) / sd for x in data] print(f"Z-scores: {z_scores}") # Find probability value is <= 55000 prob = norm.cdf(55000, loc=mean, scale=sd) print(f"Prob(X <= 55000): {prob}") # Find the value at a specific percentile value_95th = norm.ppf(0.95, loc=mean, scale=sd) print(f"95th percentile value: {value_95th}")
```

WHY Is Normal Distribution Important?

1. **Most real data is normally distributed.** Height, weight, test scores, errors, variations — they all follow this pattern.
2. **Statistical tests assume normality.** Many tests (t-tests, ANOVA) require your data to be normally distributed to work properly.
3. **Outlier detection.** You use the 68-95-99.7 rule to flag unusual values.
4. **Predictions & forecasting.** If you know data is normal, you can estimate ranges for future values.

WHERE Do You Use Normal Distribution in Your Job?

- **Anomaly detection:** Flag transactions that are 3 SD away from normal (likely fraud)
- **Quality control:** Check if product dimensions are within normal range
- **Forecasting:** 'Next month's sales should be 50K $\pm$ 5K with 95% confidence'

- **Risk analysis:** Understand how likely extreme events are
- **A/B testing:** Many tests assume normal distribution to calculate p-values

■ ■ Important: Checking if Your Data is Normal

NOT all data is perfectly normal. Before using statistical tests, you should **check** if your data looks normal.

How to Check (3 Methods):

1. Visual Inspection: Create a Histogram

Plot your data in a histogram. Does it look like a bell? Good! Is it skewed left or right? Might not be normal.

2. Check Mean = Median

If mean and median are very close, data is likely normal. If they're far apart, data is skewed.

3. Statistical Test: Shapiro-Wilk Test

```
from scipy.stats import shapiro data = [45000, 48000, 50000, 52000, 55000, 60000, 35000, 49000]
statistic, p_value = shapiro(data) if p_value > 0.05: print("Data looks NORMAL! (p-value > 0.05)")
else: print("Data might NOT be normal (p-value < 0.05)")
```

⇒ ■ Practice Problems (Normal Distribution):

Problem 1: Customer order values: Mean = Rs 1,000, SD = Rs 100. An order for Rs 1,300 is how many SDs above the mean? Is this unusual?

Problem 2: Test scores are normally distributed: Mean = 70, SD = 10. Using the 68-95-99.7 rule, what percentage of students scored between 60 and 80?

Problem 3: Daily revenue: Mean = Rs 100,000, SD = Rs 10,000. A day with Rs 70,000 revenue is how many SDs below the mean? (Calculate the z-score)

Quick Reference Cheat Sheet — Tuesday

Concept	Formula/Rule	When to Use	Data Analyst Use Case
Probability	Favorable / Total outcomes	Any time you need likelihood	Churn probability, fraud risk, model outputs
Normal Distribution	Bell curve: Mean = Median = Mode	Before applying statistical tests	Anomaly detection, forecasting ranges
68-95-99.7 Rule	Within 1/2/3 SD: 68%/95%/99.7%	Finding what's normal vs unusual	Detecting outliers, setting thresholds
Z-Score	$z = (\text{value} - \text{mean}) / \text{SD}$	Standardizing data, comparing distributions	Abnormal scoring, feature scaling
P(A AND B)	$P(A) \times P(B)$	When both events must happen	Joint probabilities in predictions
P(A OR B)	$P(A) + P(B)$	When either event could happen	Risk of any problem occurring

■ Key Takeaways from Tuesday:

- 1. **Probability is everywhere:** Every prediction, every model, every decision involves probability.
- 2. **Normal distribution is nature's favorite pattern:** Learn the 68-95-99.7 rule and you'll understand most real-world data.
- 3. **Z-scores standardize data:** They let you compare different datasets on the same scale.
- 4. **Outliers are easy to spot:** Anything 3+ SDs away from the mean is extremely unusual and worth investigating.
- 5. **Always check if data is normal:** Before applying statistical tests, visually inspect or run Shapiro-Wilk test.

Next Steps:

Tomorrow (Wednesday), we'll learn **Correlation & Covariance** — how to find if two variables move together. This is crucial for feature selection in machine learning!